



Date: 21st May 2024

TO WHOM IT MAY CONCERN

This is to state that **Savul Hameed Sahuran** Student Admission Number: (CC-MT/AIR-B001/SR-2023) has successfully completed the course **Artificial Intelligence and Robotics** at our institution.

The course contents are as follows:

Introduction to Programming

- Introduce Programming Language
- Installation of Pycharm and Python IDLE
- First Code in Python
- Variables in Python
- Python Project and Assignments

Mobile Application Development without Coding

- Introduction To MIT App Inventor
- How To Use Palette and Components
- Designing The App Layout Structure
- Developed The Code by Using Block Programming
- Connecting Firebase Database to the App
- Install the Developed App to an Android Mobile by using AI Companion or Emulator

Robotic Simulation

- What Is Simulation
- Introduction to Vexvr
- Designing the Drawing Robot
- Developed the Robot by Using Block Programming
- Testing the Robot

IOT (Internet of Things) Smart Technology

- Fundamental of Electronic
- Introduction to Arduino (Functions and Programming)
- Motion Sensor Project in Arduino
- Smoke Detection with Fire Alarm Project in Arduino
- Moving Object Detection Project with Ultrasonic Sensor in Arduino
- Temperature Detection Project in Arduino
- IR Remote Control Project in Arduino
- Complete Smart Home Project in Arduino

Making Obstacle Avoiding Handmade Robot by Using Arduino

- Equipment needed to build a robot
- Functions of Every Equipment
- Understanding the Diagram of the Robot
- Build the Obstacle Avoiding Robot
- Programming to the Robot
- Testing and Debugging the Obstacle Avoiding Robot

We wish success in all future endeavors....!

Thank you.

M.I.M. Ramseen
Managing Director